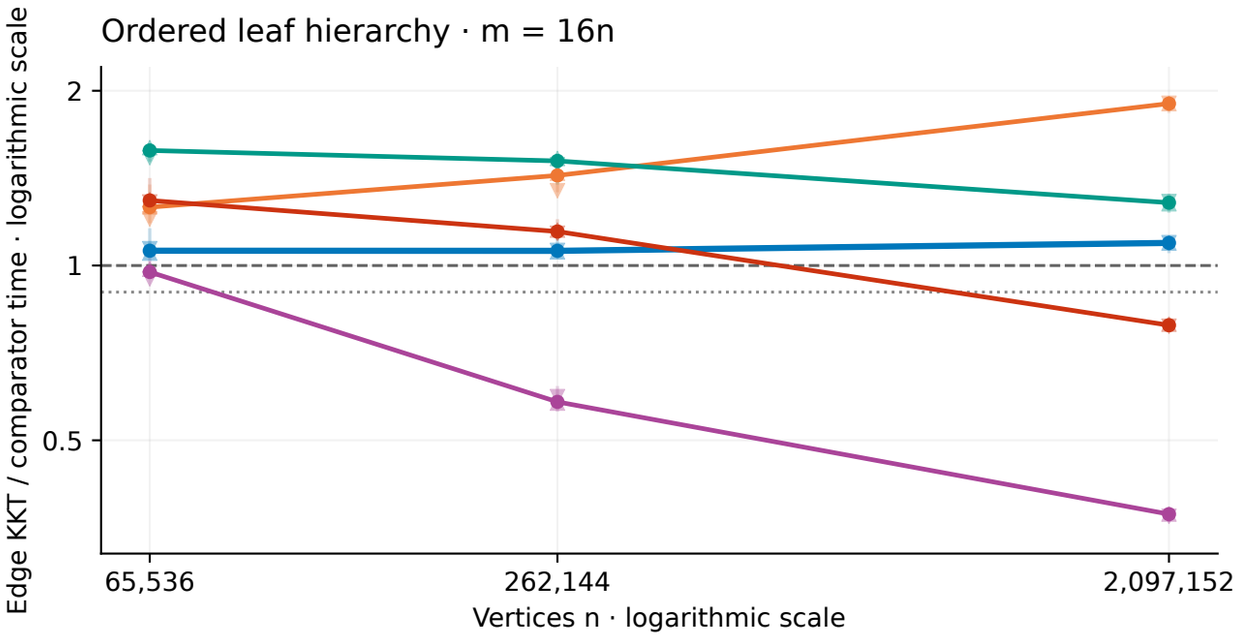
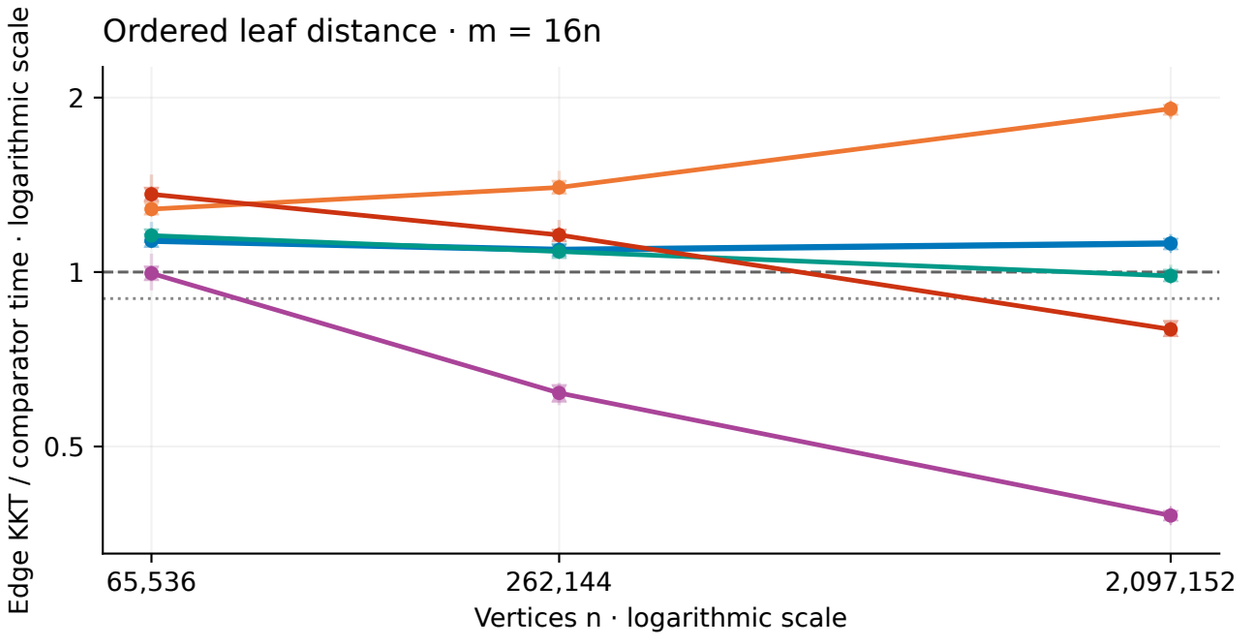
Direct Edge KKT against all five MST comparators
Process CPU time

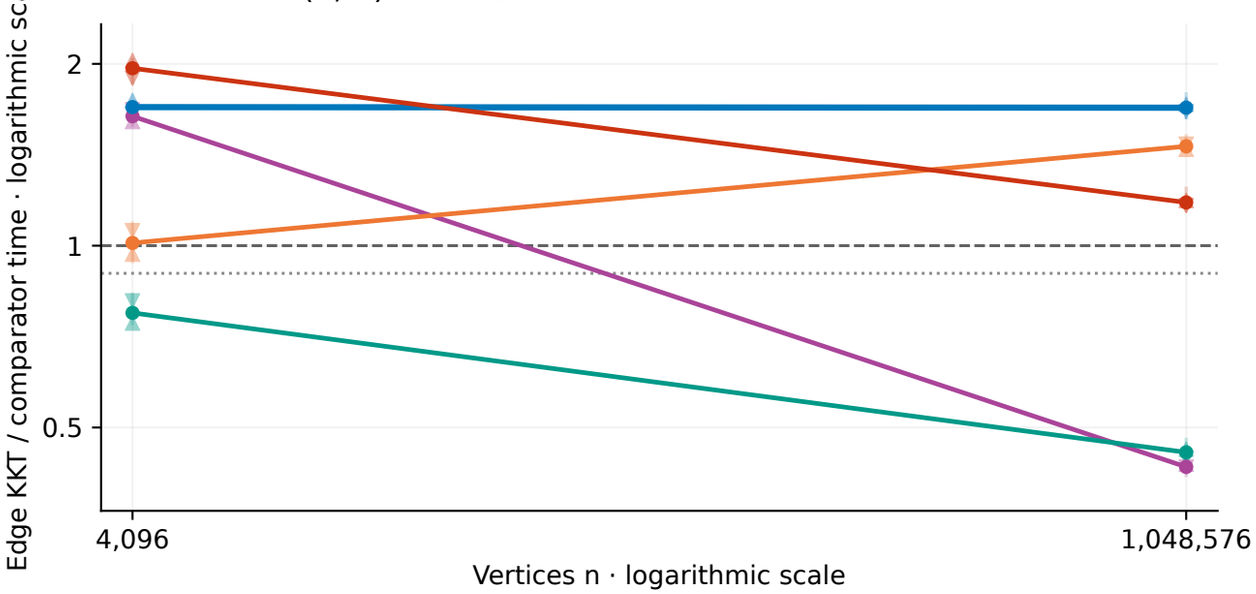
Ordered leaf hierarchy · m = 16n



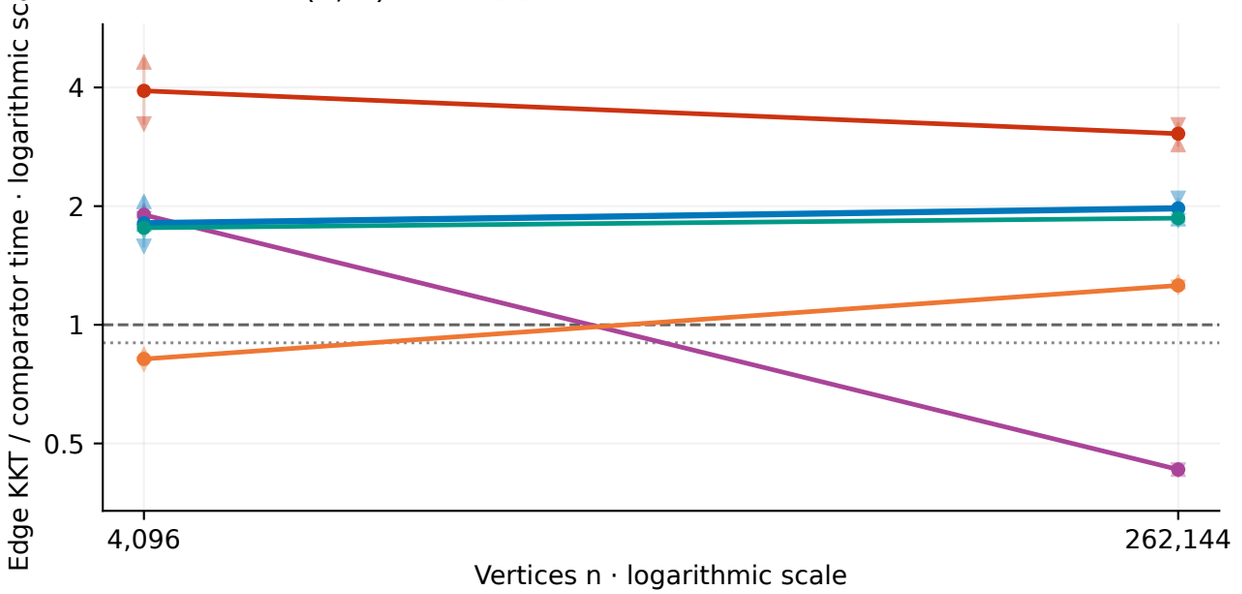
Ordered leaf distance · m = 16n



Connected G(n,m) · m = 16n



Connected G(n,m) · m = 128n



Compacting Borůvka
Retained-edge Borůvka
Kruskal
Prim
Original Borůvka
Graph medians (two seeds)
Full trial range

KKT/comparator pairs share a graph and repetition in the same Edge child. Below 1 favors KKT; dotted line: 0.90.
Solid points: median of all 12 paired ratios; faint triangles: six-trial graph medians. All cases, losses and full trial ranges retained.
Ranges are descriptive, not confidence intervals. Lines connect tested sizes; no threshold interpolation. All 0 flagged calls remain included.